Predicting Verification Latency for Post-Quantum Archival Evidence

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Abstract

Verification costs in archival evidence systems depend on parsing and validation as well as cryptographic operations. An earlier implementation reconciled its byte ledger but underpredicted latency and reused a result variable inside the timer. This study freezes a validation-aware affine model of complete-query verification latency, covering fixed overhead, record validation, signature checks, enrollment objects and Merkle hashes, before testing its predictions on fresh, larger archives. Three evidence designs were tested at zero, two and five renewals. Fresh development used 72 histories at 1,000 and 10,000 records. The code, protocol and fitted models were frozen and a private hosted snapshot was saved before 90 evaluation histories at 100,000 records. Evaluation produced 1,080 measurements across balanced persistent-process and isolated-process modes. The corrected model achieved 18/18 conditions within 15% median relative error in balanced mode and 18/18 in isolated mode. Its balanced-mode median absolute relative error was 1.63%, compared with 39.86% for the bytes-only baseline. The prespecified balanced-mode support rule was satisfied. All fresh archive and query ledgers reconciled and all 233 verifier qualification checks passed. These results evaluate extrapolation from 10,000 to 100,000 records in one warm-cache Linux implementation. The earlier failure informed the revision protocol; fresh archives test the frozen correction.

Keywords: archival evidence, post-quantum signatures, verification latency, cost prediction, reproducible benchmarks, Merkle trees

1 Introduction

Long-lived evidence must remain verifiable as signing keys and algorithms change. Evidence Record Syntax and its XML form already describe renewal chains and tree-based aggregation [1], [2]. Post-quantum signature standards provide additional choices for signing those records [3], [4]. The operational question addressed here is narrower: can a model calibrated on small archives predict the latency of verifying a complete request against a larger archive? Signature counts alone do not capture verification latency. The verifier also parses evidence, validates canonical manifests, binds the proof to the caller's

request, checks enrollment and predecessor links, reconstructs the tree path and applies the historical eligibility policy.

The earlier VITAL Q campaign exposed this gap. Its component model had a median absolute relative error of 44.70% across 18 evaluation conditions, compared with 26.27% for a bytes-only comparator. All 18 signed errors were negative. A separate measurement review found that assigning the next result to an existing variable occurred before the timer stopped. The present revision treats those data as development history and replaces their performance claims with fresh measurements.

This study contributes a prospectively frozen, validation-aware model of complete-query verification cost, tested at a tenfold increase in archive size against equally trained baselines and assessed for whether the result persists when each query runs in a fresh process. A corrected timer and balanced query order define the measurement protocol. The cryptographic primitives, batching method and renewal concept are established techniques. The manuscript makes no claim to a new cryptographic construction.

2 Related work

ERS aggregates evidence using hash trees and renews archival timestamps as algorithms or keys approach the end of their usable life [1]. XMLERS expresses related structures in XML [2]. This prototype borrows the general renewal and aggregation pattern but uses its own fixed-width envelope. It is not an ERS or XMLERS conformance implementation and its simulated clock does not supply a trusted archival timestamp.

FIPS 204 standardizes ML-DSA, while FIPS 205 standardizes SLH-DSA [3], [4]. Their distinct algorithm families motivate the parallel-branch design used in this benchmark. Post-quantum signatures use liboqs through its Python bindings and ECDSA uses cryptography. The Open Quantum Safe project describes liboqs as a tool for prototyping and evaluating post-quantum cryptography [5].

Kao’s preprint studies post-quantum audit evidence, migration patterns and Merkle-root anchoring [6]. The individual Wathiqa Internet-Draft proposes an algorithm-agile evidence-record profile [7]. These works describe post-quantum renewal and aggregation for evidence preservation; they do not report the prospective, complete-query latency extrapolation evaluated here. PROV-DM supplies a broader vocabulary for artifact provenance [8]; signed manifests can protect recorded references while leaving the truth of source acquisition outside the verifier’s scope.

3 Materials and methods

3.1 Evidence designs and verification contract

Each archive contains n canonical 1,024-byte manifests. A manifest records an identifier, payload digest, code digest, configuration digest, environment digest and provenance fields. Payload bytes are stored separately. The batch capacity is 4,096 records. The study evaluates R = 0, 2 and 5 renewals, so each checked history contains G = R + 1 generations.

Table 1 Evidence designs and verification work

Design	Stored structure	Full token checks	Selective token checks
N	Individual record chains	nG	G
E	Batched Merkle chains	kG	G
D	Two batched signature families	2kG	2G

Here k = ceil(n/4096). N and E begin with ECDSA P-256 and use ML-DSA-65 for subsequent generations. D adds a parallel SHA-384 tree and SLH-DSA SHA2-192f branch. Later key rotations within a branch retain that branch's signature family; this does not constitute hash-family renewal. The verifier accepts the requested checkpoint, query mode and record identifier as external inputs. It rejects proofs for a different request even if their internal signatures are valid. A full audit checks all manifest and token evidence required by the selected checkpoint. A selective query checks one requested record and its inclusion path where required. The trust model assumes honest public-key enrollment and a modeled local clock. The policy checks whether an eligible branch spans the requested history. One healthy branch can satisfy the D policy after a modeled failure of the other signature family. Compromised acquisition, dishonest enrollment, shared administrative failures, simultaneous failure of both branches and hash-family failure remain outside that claim.

3.1.1 Trusted enrollment and archival trust

Trusted enrollment is an external assumption: the caller supplies the expected public keys and their authorized association with each history, branch and generation. Verifying agreement with that policy does not establish the enrolling party's real-world identity or the honesty of enrollment. An attacker who controls the trusted policy is outside the qualification claim.

The modeled epoch and generation times test ordering and eligibility. A signature authenticates the asserted time value; it does not establish when the event occurred. RFC 3161 requires a timestamp authority to use a trustworthy time source [12]. This prototype supplies no such timestamp evidence.

Verification correctness means that the evidence satisfies the implemented request, structure, signature and policy checks under these assumptions. Long-term archival trust additionally depends on trustworthy acquisition, key custody, authentic enrollment, retained validation evidence and timely renewal. ERS identifies certificates, revocation information, trust anchors and policy details as relevant verification material [1]. The 233 qualification checks do not establish these operational assurances. Distinct signature families in D do not make shared enrollment or administrative failures independent and a surviving eligible branch cannot repair compromised source evidence.

3.2 Diagnosis and measurement correction

In the original helper, the statement result = verify_proof(...) could allow disposal of the previously returned result during the next timed call. The revised helper assigns the return value to a fresh local variable, stops the timer immediately after verification returns and performs assertions and disposal afterward. A regression test records destructor and clock events and requires disposal to follow the stop clock for every call. A second test requires

the previous returned object to be absent before the next call. The regression tests establish the revised timer boundary; they do not quantify how much of the earlier prediction error was caused by result disposal. The ordinary verifier returns counts and eligibility flags with an empty manifest list.
Both revised modes sequentially warm each proof file and perform explicit garbage collection before its timed verification. The same untimed cryptographic primer runs once at worker startup. In balanced mode a worker persists across the six query tasks for one history. In isolated mode each query uses a fresh worker. Process startup, interprocess communication, priming, proof construction, proof hashing, cache warming, explicit garbage collection, result assertions and returned-result disposal are outside the timer. Routine verifier internal cleanup is inside it. These endpoints measure warm-cache verification, not end-to-end request service time.

3.3 Fresh collection and fixed evaluation protocol

Table 2 Fresh version 4.2 collection

Phase	Records	Matched blocks	Histories	Timed observations
Development	1,000 and 10,000	12 per size	72	864
Evaluation	100,000	30	90	1,080

Each matched block contains N, E and D archives with matching synthetic manifests and the same selected record. Signing keys and history identifiers are fresh. Six query tasks combine the three checkpoints with full and selective verification. A six-row Williams design [13] makes each task appear equally often in each position and balances all 30 ordered predecessor pairs. Development repeats the design twice at each size; evaluation repeats it five times. The six arm permutations are also balanced. Timing-mode order is counterbalanced, with 15 evaluation blocks beginning in each mode.

Version 4.2 recollected calibration, development and evaluation on an AMD EPYC 9V74 host after the Intel host became unavailable. The 31 completed Intel histories and 372 observed measurements were not recovered and were excluded from all version 4.2 analyses. Model forms, fitting objective, workload grid and acceptance rule were preserved. The experiment uses one measurement process at a time on a virtualized Linux host. All mutable files reside in a private temporary execution directory during collection and are saved after each completed history, while all measurement workers are stopped. The next history begins only after a successful durable save.

Earlier revision attempts had two pre-evaluation infrastructure failures: an import notice disrupted JSON communication before any timing row was recorded and a later archive-size mismatch interrupted a provisional development run. The latter coincided with deleted temporary files reappearing through workspace synchronization. Those provisional development measurements were preserved in the audit folder and excluded from the current fit and evaluation. No evaluation data existed when those two failures occurred.

Before the fresh 100,000-record evaluation, the code, schedule, model definitions, fitted coefficients, acceptance rules and development-input hashes were frozen in a private hosted

snapshot. The previous campaign and evaluation scale were known when this protocol was set. The snapshot documents this revision prospectively; it is not public preregistration or independent attestation of the host clock.

The snapshot service records creation at 2026-09-25T16:59:44.190665Z. The returned metadata and snapshot digest are retained in the audit. This service date can be compared with the reported local evaluation interval; it does not independently attest the host clock.

The recorded runtime used Python 3.12.14, liboqs-python 0.16.0, cryptography 46.0.0, NumPy 2.3.5 and SciPy 1.17.0. Native library hashes and the complete platform record are included in the package. Storage uses Linux fsync calls [9]; the benchmark records their software returns and does not test physical power-loss persistence.

3.4 Corrected model and fair comparators

The corrected model predicts verification time as an affine sum of validated workload counts. Its seven nonnegative coefficients are fitted jointly from complete balanced-mode development queries:

$$\hat{T} = \alpha + \rho m + \beta_E v_E + \beta_M v_M + \beta_S v_S + \kappa e + \eta h \quad (1)$$

Here α is fixed query overhead; m is the number of validated manifests; vE, vM and vS are ECDSA, ML-DSA and SLH-DSA token checks; e is the number of enrollment objects; and h is the number of internal Merkle hashes, including selective-path hashes. A full query has m = n and a selective query has m = 1. This model is an empirical allocation of complete-query work. Its coefficients do not separately identify the causal time of individual parser or signature operations and SHA-256 and SHA-384 internal hashes share one fitted term.

For each of 36 development conditions, the fitting procedure takes the median latency and feature vector across 12 histories. Nonnegative least squares, implemented with scipy.optimize.nnls [14], minimizes squared relative residuals by dividing each design row and target by that condition's median latency. This prevents multi-second full audits from dominating the sub-millisecond selective conditions. Column normalization is undone after fitting. The complete normalized design has full rank; rank and condition number are reported with the coefficients.

Five comparators use the same development cells and the same weighting objective: bytes only; bytes with an intercept; tokens and bytes with an intercept; the corrected model without its intercept; and the corrected model without record validation. A separate 23-feature primitive model is recalibrated through ten independent microbenchmark blocks and retained as a diagnostic of the earlier modeling approach. Because that comparator uses different training evidence, it is not used to isolate model-form superiority. Coefficients remain unchanged during evaluation, including the isolated-mode sensitivity analysis.

3.4.1 Calibration overhead and deployment requirements

Calibration requires representative complete-query measurements followed by a nonnegative coefficient fit. The seven-term model uses 432 balanced development observations aggregated into 36 cell medians and a 36 × 7 fitting matrix. Isolated-mode observations and the ten-block primitive microbenchmark calibration do not train this fit.

Calibration-only collection time, solver time and solver peak memory were not separately reported; no measured calibration-runtime estimate is therefore claimed.

Once workload counts are available, prediction requires six coefficient-count products plus the intercept and their sum. Obtaining the counts may itself require parsing or traversal. Deployment requires a declared timing boundary, independently checked workload counts, representative development histories, a monotonic timer, an NNLS solver and retained environment metadata. Calibration should occur outside request timing. Changes to hardware, library builds, evidence encoding or service behavior require local refitting and validation.

3.5 Endpoints and uncertainty

The signed error for a condition is (median predicted time − median observed time) / median observed time. A condition passes when its absolute error is at most 15%. The primary support rule requires at least 15 of 18 balanced-mode conditions to pass, at least one pass in each arm-by-query stratum and a lower median absolute relative error than the bytes-only baseline. The same rule is reported separately for isolated mode. This revision defines its criterion before its evaluation; it does not replace or reinterpret the original study's failed criterion.

Uncertainty uses 10,000 bootstrap draws with seed 2192026. Each resampled unit is a matched evaluation block; all arms, checkpoints and both timing modes remain together. Intervals are conditional on the frozen development fit, so they quantify evaluation-sample variability rather than calibration uncertainty. Cell intervals are pointwise, not simultaneous. A paired interval is also reported for the difference in median absolute relative error between the corrected and bytes-only models. The 18 conditions are a fixed workload grid, not a random sample of deployments. Failed or missing cells cannot pass and no statistical outliers are removed.

4 Results

4.1 Collection and correctness

The fresh collection completed 72 development and 90 evaluation histories, yielding 864 and 1,080 timing observations respectively. All archive bytes and observed operation counts matched their independent ledgers. The qualification suite passed 233/233 checks, including request binding, cross-library signature verification, malformed evidence, signature tampering, predecessor removal, logical crash prefixes and boundary-sized batches. The two timer regression tests passed. A final audit reverified all 486 retained selective proofs, including their enrollment policies and request binding. The frozen release remained unchanged.

Evaluation ran from 2026-09-25 16:59:52 UTC to 2026-09-25 19:02:20 UTC. The recorded interval was 122.5 minutes. This interval includes durable saves between histories; those saves are excluded from query timers. It is not a claim about total project duration or independent review. The observed maximum parent-process resident memory was 277.5 MiB; child-process peaks were not separately instrumented.

4.2 Storage accounting

Table 3 Exact archive sizes at five renewals

Records	N bytes	E bytes	D bytes
100,000	1,853,710,363	110,847,988	125,825,150

At 100,000 records and five renewals, E uses 94.02% less evidence storage than N and D uses 93.21% less. These are ledger consequences of batching in the fixed envelope. They exclude payload storage. For a selected record with 12 siblings, N and E proofs occupy 28,900 and 29,324 bytes respectively. These exact byte counts verify the storage ledger for the fixed envelope and selected proof; they do not evaluate the latency model.

4.3 Frozen model coefficients

Table 4 Frozen coefficients and their units

Term	Counted work	Seconds per unit	Microseconds per unit
α	Fixed query overhead	0.000215746687	215.746687
ρ	Validated manifests	1.51147842e-05	15.114784
β_E	ECDSA token checks	9.54181705e-05	95.418170
β_M	ML-DSA token checks	4.08771966e-05	40.877197
β_S	SLH-DSA token checks	0.00208522619	2085.22619
κ	Enrollment objects	3.59735122e-05	35.973512
η	Internal Merkle hashes	3.41325475e-06	3.413255

For α, the value is per query; the other rows are per counted manifest, token, enrollment object or internal hash. Both unit columns are rounded representations of the same frozen coefficients. The coefficients are empirical allocations from jointly fitted complete queries rather than independently measured operation costs. Displayed digits support replay and do not imply equivalent measurement precision.

The fitted design rank was 7, with normalized condition number 6.19. The fit used 432 balanced development observations aggregated into 36 cell medians. No evaluation timing entered the fit.

For example, an E full query at $n = 100{,}000$ and $R = 5$ has $k = 25$ batches and $G = 6$ generations. Its counts are $m = 100{,}000$, $vE = 25$, $vM = 125$, $vS = 0$, $e = 6$ and $h = n - k = 99{,}975$. The current trees are reconstructed once, so h is not multiplied by G. Applying the printed coefficients gives 1860.645 ms, matching Table 6. These counts must be rechecked when adapting the model to another verifier.

4.4 Prediction performance

Table 5 Evaluation comparison on the same fixed workload grid

Model	Balanced passes	Balanced MARE %	Isolated passes	Isolated MARE %
Validation aware affine	18/18	1.63	18/18	2.55
Without intercept	13/18	6.96	13/18	6.71
Without record validation	10/18	13.79	9/18	13.81
Bytes only	6/18	39.86	6/18	39.23
Affine bytes	8/18	19.87	8/18	20.66
Affine tokens and bytes	8/18	16.93	8/18	17.71
Primitive components	5/18	41.98	5/18	41.75

MARE is median absolute relative error across the 18 fixed evaluation conditions.

The corrected model met the primary support rule. It passed 18/18 balanced-mode conditions, with a median absolute relative error of 1.63% and a maximum of 9.67%. The bytes-only baseline passed 6/18, with median absolute relative error 39.86%. The separately calibrated primitive model passed 5/18 and had median absolute relative error 41.98%.

The validation-aware affine model had the lowest median error among the six equally trained models (Table 5). This descriptive ranking is separate from the prespecified paired comparison against bytes only.

The paired difference in median absolute relative error (corrected model minus bytes-only baseline) was -38.23 percentage points, with a conditional 95% interval from -38.63 to -36.50 percentage points. Negative values favor the corrected model. The no-intercept and no-record-validation ablations in Table 5 show the contribution of these terms under joint refitting; they are not causal decompositions of execution time.

Table 6 Corrected model in balanced mode

Design query	R	Observed ms	Predicted ms	Error %	95% error interval %
N selective	0	0.363	0.362	-0.33	-6.78 to 7.64
N selective	2	0.559	0.516	-7.62	-12.16 to -4.29
N selective	5	0.736	0.747	1.42	-4.83 to 4.14
N full	0	11000.257	11053.547	0.48	-0.46 to 1.36
N full	2	18880.617	19229.058	1.85	0.87 to 2.42
N full	5	30779.241	31492.325	2.32	1.85 to 2.74
E selective	0	0.390	0.403	3.27	0.55 to 9.44
E selective	2	0.617	0.557	-9.67	-15.02 to -3.94
E selective	5	0.753	0.787	4.57	1.30 to 8.55

Design query	R	Observed ms	Predicted ms	Error %	95% error interval %
E full	0	1792.585	1855.356	3.50	2.08 to 4.44
E full	2	1793.628	1857.472	3.56	0.45 to 3.92
E full	5	1769.133	1860.645	5.17	4.03 to 5.83
D selective	0	2.592	2.565	-1.02	-7.12 to 2.91
D selective	2	6.990	6.961	-0.41	-3.93 to 2.32
D selective	5	13.645	13.556	-0.66	-2.01 to 2.65
D full	0	2244.778	2248.763	0.18	-0.99 to 1.31
D full	2	2347.490	2355.212	0.33	-0.46 to 1.90
D full	5	2483.132	2514.885	1.28	0.48 to 1.91

Table 7 Corrected model in isolated mode

Design query	R	Observed ms	Predicted ms	Error %	95% error interval %
N selective	0	0.353	0.362	2.50	-1.76 to 13.72
N selective	2	0.530	0.516	-2.61	-9.24 to -0.93
N selective	5	0.682	0.747	9.45	1.45 to 13.15
N full	0	10967.980	11053.547	0.78	0.10 to 1.50
N full	2	18915.109	19229.058	1.66	0.80 to 2.63
N full	5	30874.936	31492.325	2.00	1.40 to 2.54
E selective	0	0.356	0.403	13.30	4.10 to 17.37
E selective	2	0.541	0.557	3.01	-6.50 to 7.19
E selective	5	0.727	0.787	8.35	-2.83 to 15.28
E full	0	1780.315	1855.356	4.22	2.21 to 5.84
E full	2	1775.187	1857.472	4.64	3.37 to 5.49
E full	5	1787.028	1860.645	4.12	2.29 to 5.90
D selective	0	2.529	2.565	1.43	-3.57 to 4.05
D selective	2	6.747	6.961	3.18	-2.11 to 4.51
D selective	5	13.537	13.556	0.14	-3.28 to 3.62
D full	0	2215.664	2248.763	1.49	1.07 to 2.48
D full	2	2359.313	2355.212	-0.17	-0.52 to 1.47
D full	5	2475.881	2514.885	1.58	-0.61 to 1.97

Tables 6 and 7 each report 30 matched histories per row. Intervals are pointwise and conditional on the frozen model. A point pass requires an absolute error of at most 15%;

interval containment is reported in the accompanying CSV and is not the primary acceptance rule.

4.5 Process isolation sensitivity

Without refitting, the corrected model passed 18/18 isolated-mode conditions and had median absolute relative error 2.55%. The isolated-mode support rule was satisfied. Paired isolated-to-balanced latency ratios and predecessor summaries are included in the package. Mode order was counterbalanced; each balanced task appeared five times in every position and each ordered predecessor pair appeared five times per arm.

Entire pointwise 95% error intervals lay within the 15% tolerance in 17/18 balanced conditions and 16/18 isolated conditions. The primary criterion uses point estimates; it does not require interval containment.

Across the 18 conditions, the medians of paired isolated-to-balanced ratios ranged from 0.902 to 1.009. These ratios measure sensitivity to fresh-process execution under the revised cache and priming rules. They do not identify which individual allocator, cache or library state caused any difference, nor do they prove that the old timer defect caused the earlier order dependence.

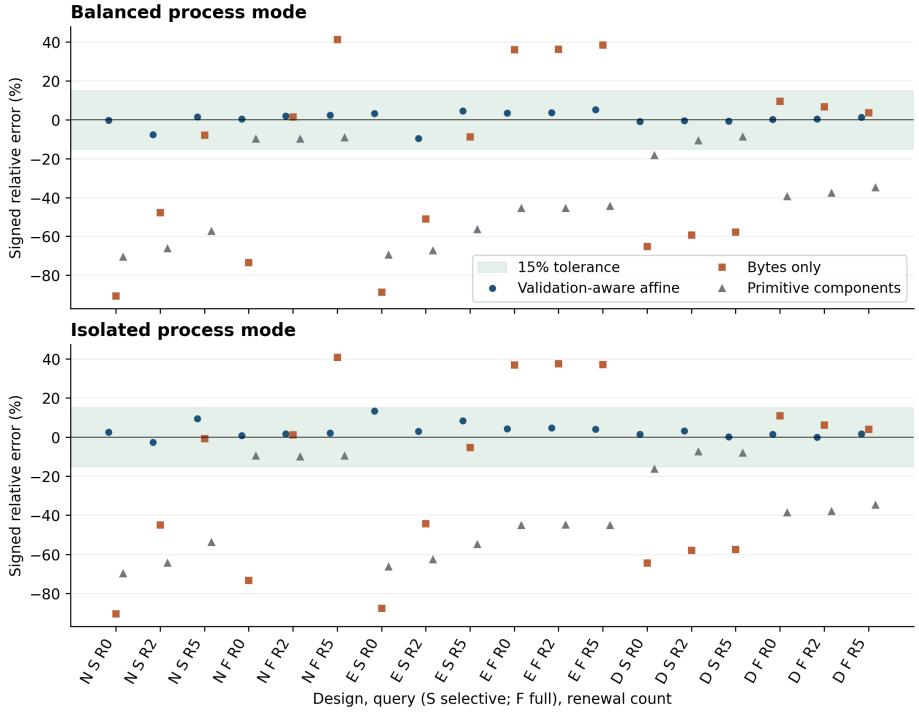


Fig. 1 Signed prediction errors across all 18 conditions. Shading marks the prespecified ±15% tolerance. The figure uses saved evaluation results.

4.6 Secondary artifact integrity example

The retained MHEALTH example is limited to functional artifact checks. The example uses the public MHEALTH dataset [10], whose associated introductory publication describes

mHealthDroid [11]. The retained workflow contains 5,000 selected sensor windows and five derived pipeline artifacts. The original package records 12/12 successful mutation checks across N, E and D: changed payload bytes are detected by the separate payload audit, while altered signed references and missing predecessors are rejected by evidence verification. These are historical functional fixtures and do not enter the fresh timing analysis. Classification accuracy, stair-climbing recall and subject-level generalization are omitted as evidence for the latency question. There is no claim of clinical utility, device validation or authenticated physiological acquisition.

5 Discussion

The revised experiment separates accounting correctness from timing prediction. Byte and operation ledgers can be exact while their cost coefficients omit work or aggregate it at the wrong level. Complete-query fitting incorporates canonical manifest validation and fixed setup costs that a primitive-only model can miss. The comparison with equally trained bytes-only and affine baselines makes the predictive claim assessable; the ablations test whether the additional terms are useful within this workload.

Under the declared criterion, the corrected model met the primary performance requirements in balanced mode. Its isolated-mode result must be read alongside that finding because a capacity estimate can change when process reuse changes. A practical use is to estimate verification time for the same envelope, library build and cache profile after local calibration. The results do not establish portability to another parser, signature implementation, batch size, cold-cache service or storage system.

The timer fix is necessary for a clean measurement boundary, but the study does not attribute every improvement to that fix. The model form, fitting evidence, weighting, process priming, garbage collection and query schedule also changed. Thus the experiment evaluates the revised pipeline as a whole. A causal estimate of the timer defect alone would require a separately controlled intervention with all other conditions held fixed.

The contribution is empirical and methodological. Tree batching and signature renewal explain the storage reduction without requiring a new theory. The additional evidence comes from testing whether a workload-aware affine model, frozen after development, transfers to ten times the largest development archive and to a separately reported process-isolation mode. The resulting claim concerns measured prediction accuracy for this implementation and workload.

5.1 Limitations

Development and evaluation used different archive sizes in sequential phases, so scale and collection time are not independently randomized. One virtualized host cannot establish cross-platform performance, dedicated-host reproducibility or tail-latency service guarantees. Each history contributes one timed observation per condition and mode. Bootstrap intervals are conditional on the fitted coefficients and pointwise across the fixed grid. They omit development-fit uncertainty and do not provide simultaneous coverage.

The protocol was informed by the earlier failure and used a previously exercised evaluation scale. The preserved private snapshot strengthens the record of what was fixed before the new measurements, but it is neither public preregistration nor a trusted timestamp. Independent replication is needed to assess performance beyond this execution environment. The exact source of the interrupted workspace archive-size mismatch was not independently diagnosed; all provisional development data were excluded.

The declared retention rule discards full archives and full query proofs after their measurements to bound disk use. The approximately 4 GiB retention limit kept metadata, logs, model inputs and 486 selective proofs but did not retain every generated archive and proof. It originated under this allowance and was carried forward on the current host. The saved materials support reanalysis and direct reverification of the retained subset; regenerating discarded material permits a new execution but cannot recover the original randomized signatures for complete historical reverification.

The security evaluation covers explicit adversarial fixtures under honest enrollment and a modeled clock. Physical durability, source authentication, common-mode failures and production hardening remain untested. The MHEALTH example inherits the limitations of secondary public-data reuse and is not a clinical study.

5.2 Generalization and adaptation

The calibration procedure and explicit accounting of verification work can be adapted to other systems. The numerical coefficients are specific to the tested AMD EPYC host, cryptographic builds, fixed-width envelope and warm-cache endpoint. Different instruction sets, parser behavior and cryptographic optimizations can change the relative costs of individual terms; a single hardware speed multiplier is not established. Production storage may add costs absent from this measurement boundary. The current results demonstrate scale extrapolation and process-isolation sensitivity within one environment rather than cross-platform validation.

Practitioners should first define the verification request, acceptance policy and timing endpoint, then instrument the target verifier and reconcile its counts against an independent ledger. Retain the seven terms only when they describe that verifier’s work. Variable-size records, certificate-path validation, revocation processing or timestamp checks may require other terms. Collect representative development histories, fit the relative-error objective in Section 3.4 and check rank, conditioning and residuals by workload. Freeze the features, coefficients and acceptance rule before testing fresh histories. Recalibrate after material environment changes. The present median-error criterion does not establish a production tail-latency guarantee.

5.3 Cold-cache verification and network retrieval

Future evaluation should distinguish warm-cache verification, cold local retrieval and remote storage access, with explicit reporting of client, operating-system and storage caches. Restarting a worker alone does not establish cold caches. Service experiments should record end-to-end elapsed time alongside retrieval, proof-construction and verification spans, plus concurrency, transferred bytes and network conditions. Overlapping stages require tracing that accounts for overlap rather than simple addition. Warm-cache local reads already occur inside the current timer and should not be counted twice in an

extended model. These scenarios remain future work; no cold-cache or network-service performance is claimed here.

6 Conclusions

Corrected timing and fresh evaluation produced 18/18 balanced-mode and 18/18 isolated-mode conditions within the 15% error tolerance. The corrected model's balanced median absolute relative error was 1.63%, compared with 39.86% for bytes only. The declared balanced-mode support rule was met. The supported conclusion is specific to this warm-cache implementation and fixed workload grid. The byte ledger, request-binding checks and retained proofs provide reproducible correctness evidence; independent replication remains future work.

Declarations

Funding: No funding was received for this work.

Conflict of interest/Competing interests: The authors declare no competing interests.

Ethics approval and consent to participate: No new experiments involving live vertebrates, humans or human tissue samples were conducted for this study. The only human-derived data used are the publicly available, de-identified MHEALTH dataset (DOI: 10.24432/C5TW22), reused here solely for functional artifact-integrity checks rather than clinical or physiological analysis, under its original public release terms. No formal institutional ethics or IRB review was sought or required, as this study involved no new experiments on live vertebrates, humans or human tissue and relied solely on reuse of already-public, de-identified secondary data.

Consent for publication: Not applicable — this manuscript does not include any individual person's identifying data, images or videos.

Data availability: The research supplement accompanying this revision contains the existing version 4.2 development and evaluation timing rows, history records, fitted coefficients, analysis outputs, audit logs and retained selective proofs with enrollment policies. In the accompanying vital-q reproducibility repository, original timing rows, analyses, audit records and proofs are under data/published; frozen coefficients are under src/release. Full archives and full-query proofs were discarded under the retention rule described in Section 5.1. The repository is public at https://github.com/discoversaikatchakraborty/VITAL-Q; release v3 (https://doi.org/10.5281/zenodo.23329917) is the reproducibility package and release v2 adds this manuscript with the response to the reviewers. The secondary MHEALTH dataset is publicly available at https://doi.org/10.24432/C5TW22 [10].

Materials availability: Not applicable — this study did not generate or use physical laboratory materials or reagents.

Code availability: The research supplement contains the existing verifier, archive builder, timing helpers, calibration, model fitting and analysis code under src/release in the accompanying vital-q repository. The repository also includes executed analysis and control notebooks, plotting code, dependency information and instructions for numerical replay,

retained-proof verification and a fresh full-scale run. The portable driver uses local checkpoints; the original study used the hosted snapshots described in Section 3.3. A separately recorded fresh execution is included under executions/full_scale_20261012; it does not replace this manuscript's original measurements. Code and data are supplied together in the public repository https://github.com/discoversaikatchakraborty/VITAL-Q, released as v3 (https://doi.org/10.5281/zenodo.23329917) for the package with v2 adding the manuscript materials.

Use of generative AI or AI-assisted technologies in manuscript preparation: AI-assisted tools were used for implementation support, plotting and manuscript editing. Every experimental output, citation, numerical result and interpretation was checked by the authors against the underlying artifacts. AI tools were never treated as a source of empirical evidence and were not permitted to invent or alter experimental results.

Author contribution: A. Chakraborty, S. Chakraborty, S. Marik and S. Dhar jointly conceived and designed the study, developed the software and analysis pipeline, executed the data collection, performed the formal analysis and wrote the manuscript. All authors read and approved the final manuscript.

Appendix A Technical appendix

A.1 Evidence encoding and authenticated request

The version 2 envelope fixes integer byte order as unsigned big endian. Archive and proof headers are 64 bytes and identify the history, arm, generation, record count, batch capacity, modeled epoch and mode. Objects use a 16-byte frame carrying a four-byte magic, format version, type, reserved field, object identifier and body length. Verification bounds lengths before allocation and rejects unknown, duplicate, extraneous, truncated and incorrectly ordered objects. This is the study-specific evidence envelope; it is not a standards conformance claim. The full binary layout is specified in src/release/format_spec.md.

A canonical manifest is exactly 1,024 bytes of ASCII-compatible UTF-8 JSON with sorted keys, compact separators and padding inside a designated field. The verifier checks canonical serialization and ties its frame identifier to its record identifier. Each history has a random 32-byte identifier; matching synthetic manifests are identical across N, E and D within a block, but cryptographic keys and history identifiers are fresh. The externally supplied trust policy includes expected history, arm, record count, epoch and enrolled public keys. Proof contents cannot supply their own trusted keys or silently choose the requested checkpoint, query type or selected record.

The N design signs individual record subjects. E signs batches of at most 4,096 records using one primary Merkle tree per batch. D signs two independent batch branches: a SHA-256 primary tree and a SHA-384 redundant tree. Leaf hashes include a domain byte, global record identifier and canonical manifest; internal hashes use a distinct domain byte and the ordered children. Odd final nodes are promoted, not duplicated. A selected E or D proof carries only actual path siblings, while a full proof checks all required manifests and tokens

and reconstructs roots. The verifier authenticates complete predecessor chains. Missing audit objects are rejected even when another branch could satisfy the policy.

A.2 Cryptographic operations and policy boundaries

At generation zero, primary branch tokens use ECDSA P-256 with SHA-256. Subsequent primary generations use ML-DSA-65. D additionally uses SLH-DSA SHA2-192f in its redundant branch. Each generation rotates the key within a branch. The signed token binds a history prefix, branch, generation, subject, modeled issue time, target digest and predecessor digest. Public-key enrollment objects are checked against the externally supplied enrollment policy. Expected issue times follow the modeled epoch plus 100 units per generation; they are simulated policy inputs, not real timestamp authority evidence.

A policy-eligible branch must span the requested history and each requested subject must have at least one eligible complete branch. Cryptographic verification still audits all presented required branches. This separates a simulated loss of one signature family from an assertion that the remaining branch proves real-world origin or time. No new signature algorithm, hash function or recovery protocol is proposed. The retained tests exercise the protocol implementation under its stated honest-enrollment assumptions.

A.3 Ledger and timing trace

Every completed history records archive bytes, object counts, signature operations, Merkle-hash operations, barrier events and proof bytes independently of the latency model. Batch counts use k = ceiling(n/4096). With G = R + 1 generations, N requires nG signed record tokens, E kG batch tokens and D 2kG tokens. A selective N or E request checks G tokens; D checks 2G. Archive byte reductions reported in Table 3 compare evidence archives at the same n and R and intentionally exclude payload files. The selective-proof example fixes the observed path at 12 siblings and does not imply that proofs always have that length.

The measured complete-query endpoint opens an already built proof and ends at the verifier acceptance decision, including file reads, parsing, canonical validation, enrollment lookups, hashing and signatures. The timer uses perf_counter_ns. Explicit priming, proof construction and hashing, file warming, worker startup, interprocess communication, forced collection, returned-result checks and disposal occur outside it. The same cache preparation is applied to each query in each mode. A measurement is therefore a warm-cache implementation observation, not a cold storage retrieval or end-to-end network service time.

A.4 Test coverage and evaluation accounting

The retained execution records contain 162 successfully reconciled histories and a separate 233/233 verifier test record. Tests cover valid and tampered signatures, wrong keys, malformed framing, enrollment mismatch, altered requested identity, missing chain predecessors, tree boundaries and logical crash prefixes. The final audit confirms 162 archive ledgers and 486 retained selective proofs. The two timer tests separately check stop-clock order and the lifetime of the previous returned object. These counts measure implemented checks; they are not a formal security proof or a physical power-loss experiment.

The fixed evaluation grid has three designs, two query modes and three renewal counts: 18 conditions per timing mode. Every condition receives 30 matched evaluation observations. Ninety evaluation histories therefore give 1,080 timed observations, while 72 development histories supply 864. Balanced mode and fresh-process isolated mode are recorded for each history. The six-task schedule balances 36 position cells and 30 ordered predecessor pairs within each design; the audit found five observations for each relevant position and predecessor pair.

The fitted coefficients are nonnegative and use the 36 balanced development cell medians, each from 12 observations. Dividing each row by its observed median time makes the objective relative rather than dominated by large full-query times. The affine intercept captures fixed setup at this measurement endpoint; record count captures validated manifests. Signature-family counts, enrollment objects and internal Merkle hashes capture the remaining measured workload. These regressors can correlate and no coefficient should be read as a separately randomized causal operation cost. The equally trained ablations in Table 5 isolate predictive utility within the chosen model class.

The supplied raw rows, frozen model JSON, six-cell executed analysis notebook, tabular evaluation CSV and figure permit numerical replay without rerunning the roughly two-hour cryptographic collection. The notebook reconstructs the development fit and compares all evaluation cells and conditional bootstrap intervals. It does not re-create randomized signatures or lost full archives. The supplied selective proofs and policies permit rechecking that retained subset directly; exact historical full-proof verification needs bytes deliberately discarded under the retention rule.

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